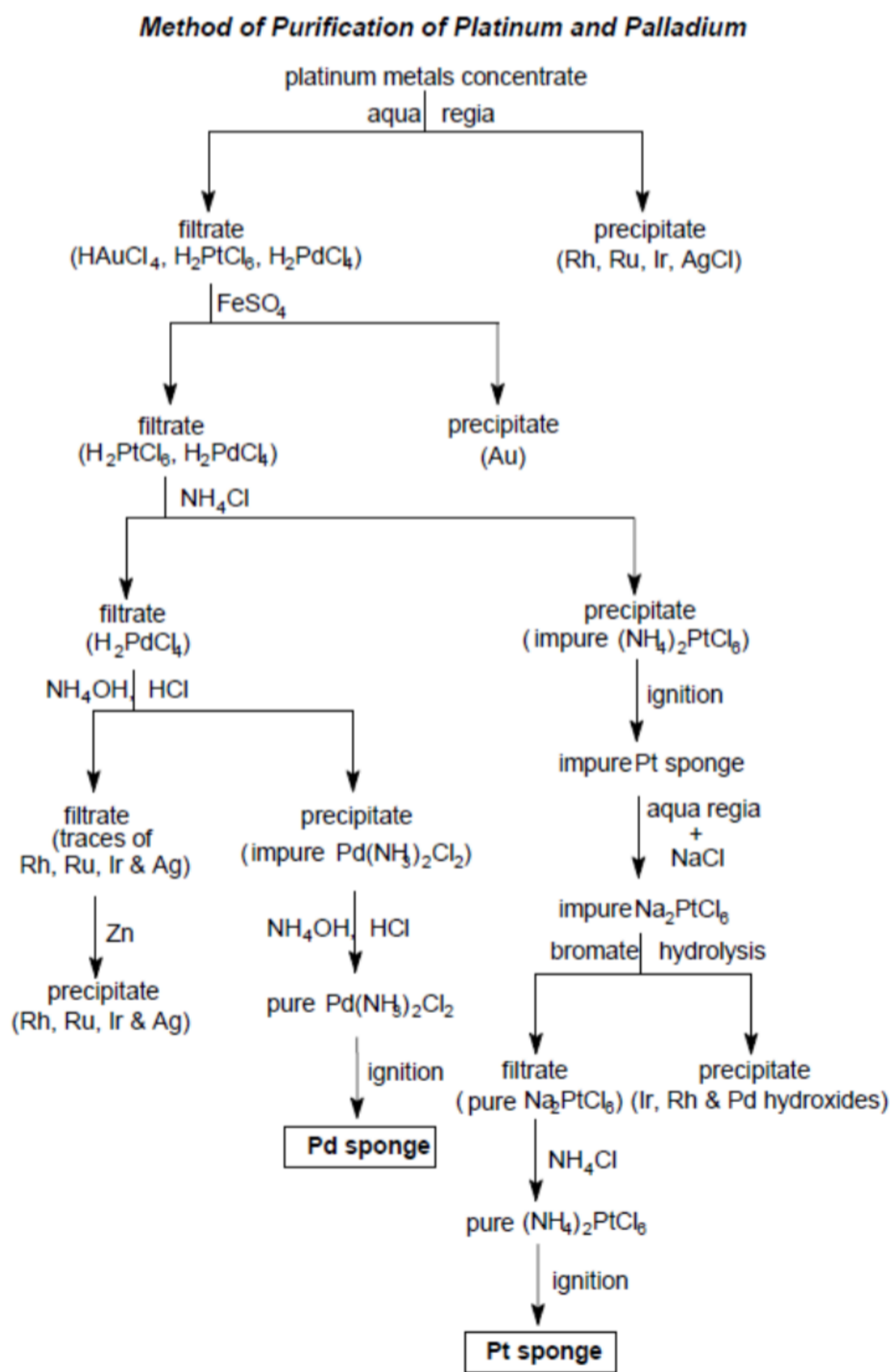


1. Much of the world's supply of platinum group metals is derived from the residues recovered from the electrolytic refining of copper and nickel. A flow chart for the recovery of platinum and palladium is shown as following.

- Clearly draw the shape (geometry) of both the PtCl_6^{2-} and the PdCl_4^{2-} anions.
- Clearly draw all possible stereoisomeric structures of monomeric $\text{Pd}(\text{NH}_3)_2\text{Cl}_2$. Label the structures that you have drawn with their correct stereochemical descriptors.
- What is the role of the FeSO_4 in the second step of the flow chart? Write a balanced equation for the reaction of FeSO_4 in this step.
- Write a complete balanced equation for the ignition of $\text{Pd}(\text{NH}_3)_2\text{Cl}_2$ in air to give Pd metal. In this reaction, what is being oxidized and what is being reduced?



2. A sample of 2.3793 g of crystallohydrate of the type $M_xA_y \cdot zH_2O$, where M is a metal, reacted with an excess of $SOCl_2$. Gaseous products formed by the reaction were introduced into a barium chloride solution containing hydrochloric acid and hydrogen peroxide. Small quantities of $SOCl_2$ carried by the gaseous products were removed by freezing out. The mass of the precipitate that is deposited from the solution, was 14.004 g. It was found to contain 13.74 mass % of sulphur.

In another experiment, 1.1896 g of the initial substance were dissolved in water and the solution was diluted to a volume of 100 mL. One fifth of this solution required to react with 10 mL of a 0.2 M $AgNO_3$ solution. The mass of the precipitate formed by the titration was 0.28664 g. (The end point of the titration was determined by the conductometric method.)

Problems:

2.1 Calculate the summary formula of the crystallohydrate.

2.2 If you know that the sample can contain a maximum of seven moles of water per one mole of the crystallohydrate, give an example of another possible hydrate that cannot come into consideration due to the given limitation.

3. A New Journey for Ancient Sulfur

Sulfur has been known and used since the ancient times. Nowadays, sulfur is widely used in chemical production and pharmaceutical industry as an inexpensive chemical raw material. Pyrite (FeS_2) is commonly used as a raw material in the industrial production of elemental sulfur. By heating pyrite in the presence of limited air supply, the theoretical yield of elemental sulfur is 100% and a black magnetic oxide (Fe_3O_4) is the other product.

3.1: Write the balanced reaction equation that describes the above conversion.

In addition to sulfur, a small amount of SO_2 is generated as a by-product during the actual process. This reaction can be monitored by measuring the amount of SO_2 . The following recipe is used for the monitoring: Raw ore powder is heated in a temperature-controlled tube furnace. The generated SO_2 is absorbed by 2 mol L^{-1} NaOH solution. When the reaction is completed, the solution is transferred to a 500 mL volumetric flask and diluted by distilled water to the mark. 25.00 mL of this diluted solution is then added into an iodine flask containing 50.00 mL 0.05122 mol L^{-1} I_2 standard solution and 5 mL 20% H_2SO_4 solution. After maintaining the iodine flask in dark for 5 min, the solution is titrated with 0.1012 mol L^{-1} $Na_2S_2O_3$ standard solution. When the color of the solution turns light brown, 3 mL 0.5% starch indicator is added. Continue the titration until the blue color disappears.

3.2: Write the balanced reaction equations involving I_2 in the above measurement.

3.3: In such a test experiment starting from pyrite, 17.6 g of elemental sulfur was collected. The analysis of the by-product gases according to the above procedure resulted in the consumption of 18.47 mL of $\text{Na}_2\text{S}_2\text{O}_3$ standard solution. Assuming that no other sulfur containing species were produced, calculate what percentage of the sulfur in pyrite was lost as the by-product?

The lithium-sulfur battery is a compelling energy storage system because its high theoretical energy density exceeds conventional Li-ion batteries. The net reaction of a lithium-sulfur battery can be simplified as: $16 \text{ Li} + \text{S}_8 \longrightarrow 8 \text{ Li}_2\text{S}$. Sulfur is the cathode and metallic lithium is the anode active material during discharge.

3.4: Write the balanced equations for reactions that take place at cathode (a) and anode (b) during discharge.

3.5: Calculate the mass ratio of the cathode active material to the anode active material, according to the net battery reaction.

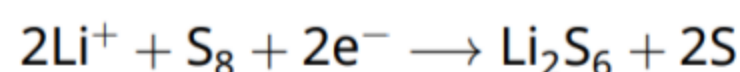
A lithium-ion battery (LIB) that possesses an average operating voltage of 3.8 V and a deliverable capacity of 3110 mAh can provide energy for a mobile phone to play videos continuously for 22 hours upon a full charge.

3.6: If the LIB is replaced by an ideal lithium-sulfur battery pack, which has an average operating voltage of 4.2 V and contains 23 g sulfur as the active electrode material that can react stoichiometrically during discharge, calculate how many hours the new battery pack will provide energy for the phone to play videos continuously after a full charge.

Elemental sulfur generally exist as S_8 molecules. In real lithium-sulfur batteries, S_8 is not directly reduced to Li_2S during discharge, but it undergoes stepwise reactions generating different soluble lithium polysulfides (Li_2S_n , $n = 3 - 8$). These lithium polysulfides could diffuse to the anode and corrode it, which results in the loss of active electrode materials. This phenomenon is called "shuttle effect".

3.7: Write the balanced reaction equation for the corrosion of anode by soluble lithium polysulfides (Li_2S_n) that produces Li_2S .

In order to suppress the "shuttle effect", the forms of polysulfides in an electrolyte have been extensively studied. Li_2S_6 is one of the most representative intermediate products:



A theoretical study shows that two conformers with comparable energies, Li_2S_6 (I) and Li_2S_6 (II) coexist in 1,2-dimethoxyethane (DME), a common electrolyte solvent in lithium-sulfur batteries. The dissociation

of Li_2S_6 in DME is shown below:

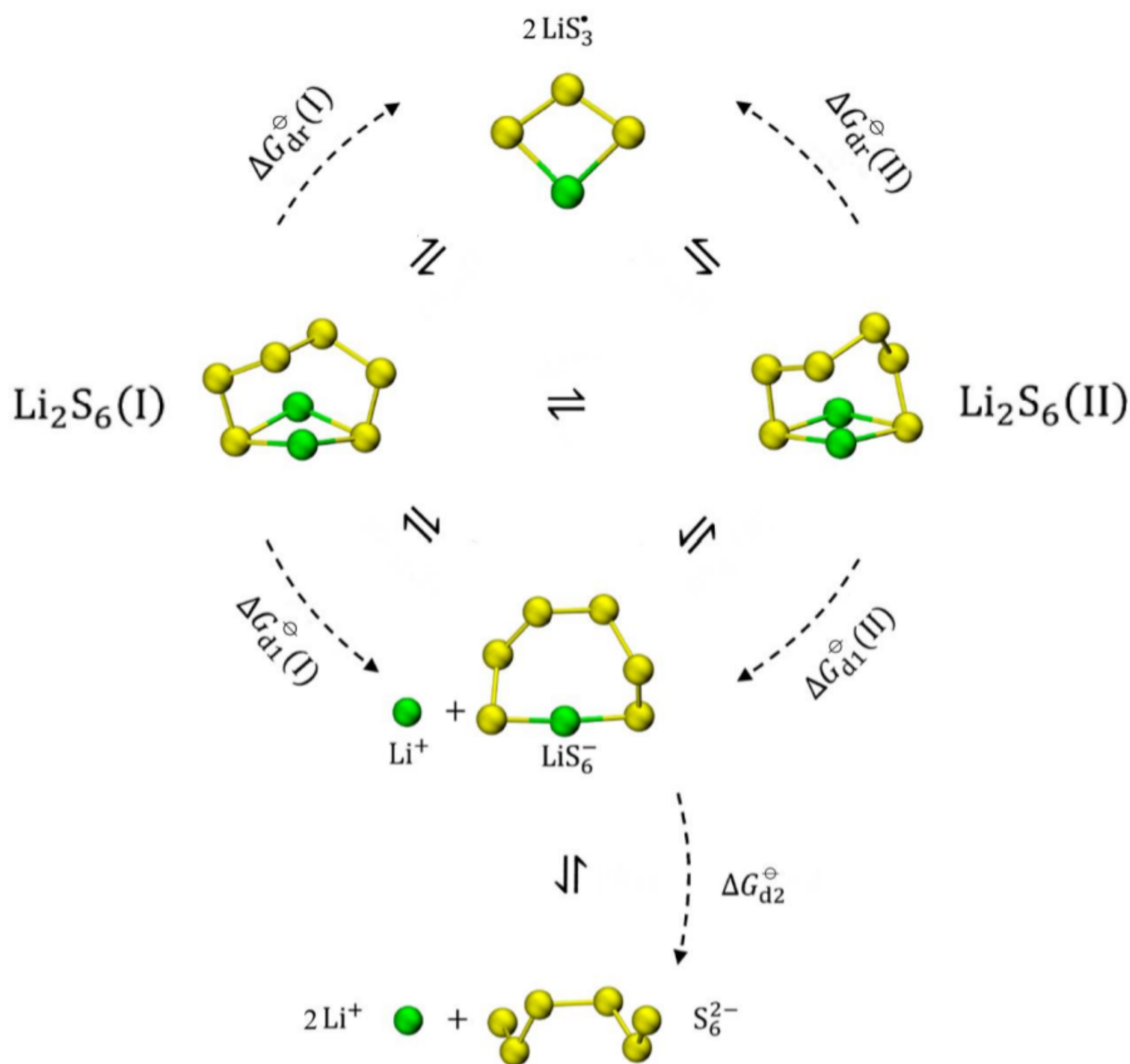


Table Dissociation Gibbs energies (kJ mol^{-1}) of different reactions in DME (298.15 K, 1 bar)

$\Delta G_{\text{d1}}^{\ominus}(\text{I})$	$\Delta G_{\text{d1}}^{\ominus}(\text{II})$	$\Delta G_{\text{d2}}^{\ominus}$	$\Delta G_{\text{dr}}^{\ominus}(\text{I})$	$\Delta G_{\text{dr}}^{\ominus}(\text{II})$
20.68	18.92	100.55	45.13	43.37

3.8: Using the data from Table, calculate the equilibrium concentration ratio of two conformers in DME

(298.15 K, 1 bar), $\frac{[\text{Li}_2\text{S}_6(\text{II})]}{[\text{Li}_2\text{S}_6(\text{I})]}$.

3.9: Using the data from Table, calculate the apparent dissociation constant of $\text{Li}_2\text{S}_6 \rightarrow \text{Li}^+ + \text{LiS}_6^-$ in DME (298.15 K, 1 bar).

3.10: Sort the following equilibrium concentrations in decreasing order: $[\text{Li}_2\text{S}_6]$, $[\text{LiS}_6^-]$, $[\text{S}_6^{2-}]$ and $[\text{LiS}_3^*]$ in DME.

The standard reduction potential of metallic lithium in water at 298.15 K and 1 bar is:

$$E^{\ominus}(\text{Li}^+/\text{Li}) = -3.040 \text{ V}$$

The standard solvation Gibbs energies of gaseous $\text{Li}^+(\text{g})$ to $\text{Li}^+(\text{sol})$ in different solvents.

	$\text{Li}^+(\text{H}_2\text{O})$	$\text{Li}^+(\text{DME})$
$\Delta G^\ominus / \text{kJ mol}^{-1}$	-116.9	-114.6

4.11: Calculate the standard reduction potential of metallic lithium electrode in DME.

Studies have shown that electromotive force of a lithium-sulfur battery would be increased by replacing DME with dimethyl sulfoxide (DMSO). Therefore, the forms of polysulfides in DMSO also attract researchers' attention.

In a test, a certain amount of Li_2S and 4.81 mg sulfur powder were added into 10.00 mL DMSO, then heated and stirred until completely dissolved (ignoring the change in volume). Suppose that only the following polysulfides were present in the DMSO: $\text{S}_3^{\bullet-}$, S_4^{2-} , S_5^{2-} , S_6^{2-} , S_7^{2-} , S_8^{2-} . The equilibrium concentration ratio of the sulfur containing species was:

$$[\text{S}_3^{\bullet-}] : [\text{S}_4^{2-}] : [\text{S}_5^{2-}] : [\text{S}_6^{2-}] : [\text{S}_7^{2-}] : [\text{S}_8^{2-}] = 17.50 : 1.00 : 4.50 : 55.00 : 5.00 : 0.75$$

4.12: Calculate the original mass m (in mg) of Li_2S added to DMSO.